3D Scene Reconstruction

Student 112006220 | Replica apartment_0 | Revised 29 October 2026. This report evaluates an RGB-D reconstruction pipeline on the supplied first-floor dataset and student-collected upstairs and downstairs routes. The main finding is that coverage and registration accuracy must be assessed separately: more observations did not guarantee a coherent map.

1. Reconstruction pipeline

Depth unprojection is implemented from scratch in NumPy. For stored integer depth D, metric depth is Z=D/s. The capture intrinsics supply width, height and horizontal FOV; fx=fy=W/(2 tan(hfov/2)), cx=W/2, cy=H/2 under square pixels. Then X=(u-cx)Z/fx and Y=(v-cy)Z/fy. Only finite positive depth is retained. No Open3D projection function is used for this required step.

The supplied capture has no depth_scale field, so integer depths use the collector convention s=1000 units/m; a recorded depth_scale takes precedence. Floating depths are treated as metres. OpenCV BGR is converted to normalized RGB. Camera axes are +X right, +Y down and +Z forward.

Each cloud is voxel-downsampled, assigned estimated normals, and described by FPFH. Mutual feature matching with geometric checks feeds RANSAC for a current-to-previous transform [1]. Open3D point-to-plane ICP refines it [2]. The pose chain is T_global = T_global @ T_current_to_previous. Copied clouds are transformed and merged; map-only downsampling does not alter estimated camera poses.

Setting	Value
Registration / merged-map voxel	0.05 m / 0.025 m
Normals	radius 0.10 m; max 30 neighbors
FPFH	radius 0.25 m; max 100 neighbors
RANSAC	3 samples; 100000 trials; confidence 0.999
RANSAC checks	edge ratio 0.9; correspondence gate 0.075 m
Point-to-plane ICP	gate 0.075 m; 100 iterations; tolerances 1e-6
Reproducibility	RANSAC seed 0; OMP_NUM_THREADS=1

2. Evaluation and visualization

Mean L2=(1/N) sum ||p_est,i-p_GT,i||, in metres. Both trajectories use the first consumed camera as reference. GT translation is R0^T(t_i-t0), with axis conversion diag(1,-1,-1) for estimated image-camera positions. Frame identities preserve pose association for subsets. No GT-based fitting, scale adjustment, initialization or registration correction is applied.

Estimated trajectories are red and GT trajectories black. Ceiling crops affect map presentation only. Optional 2 cm TSDF surface fusion uses the saved estimated poses and 8 cm truncation [4]; it does not change trajectory error. A corrected above-floor camera improves visibility but cannot repair distorted geometry.

Data Verification and Experiment Integrity

3. Measurements and workflow

Each experiment is declared before measurement. The first pass records all eight pipeline-grounded factors, then seals the declaration and machine evidence. Plots and summaries guide interventions; factors are raw observations, not inherent good/bad labels. Original evidence remains unchanged when selected runs are recorded. New captures and the temporal treatment use separate declarations.

Measurement	Scope / source
RawPointCount	Per frame: finite positive back-projected points
VoxelizedPointYield	Per frame: points after 5 cm voxel downsampling
CoarseRANSACFitness	Per pair: RANSAC inlier fraction
CoarseRANSACRMSE	Per pair: RANSAC inlier RMSE (m)
CoarseRANSACCorrespondences	Per pair: retained RANSAC correspondences
ICPInlierFitness	Per pair: refined inlier fraction
ICPInlierRMSE	Per pair: refined inlier RMSE (m)
ICPInlierCorrespondences	Per pair: retained ICP correspondences

Registration measurements are read directly from Open3D RegistrationResult [3], using the configured 0.075 m gate. Counts are direct pipeline reads. Definitions and parameter roles are preserved in docs/factors.md and ontology/hw1.ttl. For N frames, complete evidence contains 2N+6(N-1) occurrences.

Experiment	Frames	Factor records	Runs recorded
Provided floor 1	191	1522	baseline + selected
Original upstairs	388	3098	baseline + selected
Temporal upstairs	388	3098	baseline
New upstairs	738	5898	baseline
Extra downstairs	704	5626	baseline

All listed experiments pass Turtle parsing, declaration-seal and evidence-completeness checks. Report metrics are checked against recorded Turtle run values by this builder. The original two experiments include baseline and selected runs; the newer captures are baseline-only additional experiments, with no selected improvement yet established.

4. Reading evidence cautiously

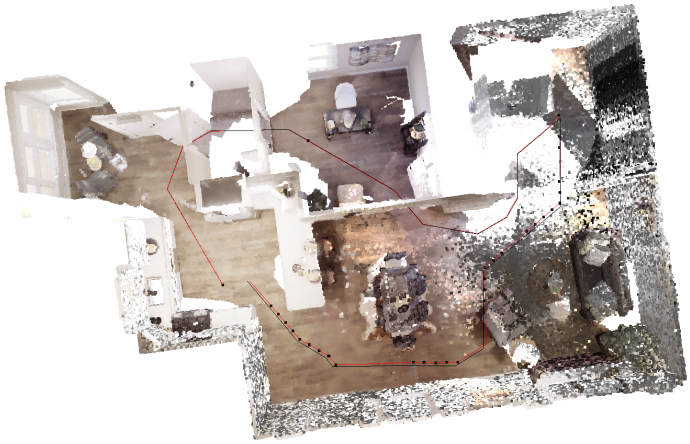
High fitness can occur on a feature-poor plane and does not prove a correct camera pose. Low inlier RMSE ignores unmatched geometry; zero RMSE with zero correspondences is a failure, not perfect registration. Point yield, pair support, RGB/depth observations, trajectory disagreement and visible map artifacts must be inspected together. Pairwise factors alone do not establish the cause of accumulated drift.

Phase 1: Provided First-Floor Dataset

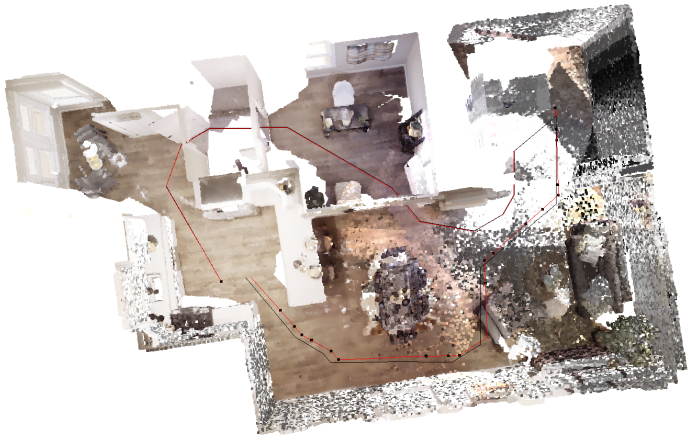
The supplied archive [5] contains 191 matched 512 x 512 RGB/depth frames and 191 poses, rather than the 152-frame example in the handout. The provided directory and observations are unchanged; the separate student downstairs capture does not replace this Phase 1 dataset.

Run	Frames	Mean L2 (m)	Elapsed (s)	Profile
Baseline	191	0.058835	371.530	standard
Selected	142	0.103037	328.458	standard

The selection query retains endpoints, odd frames, and frames with incoming baseline ICP fitness below 0.95; it removes only high-fitness even interior frames. This retains 142/191 observations with at most one omitted frame per gap. The evidence shows reduced support near frames 26-27, 107-108 and 149-151; the query is saved in the experiment queries/frames.rq.



Baseline: 191 frames



Selected: 142 frames

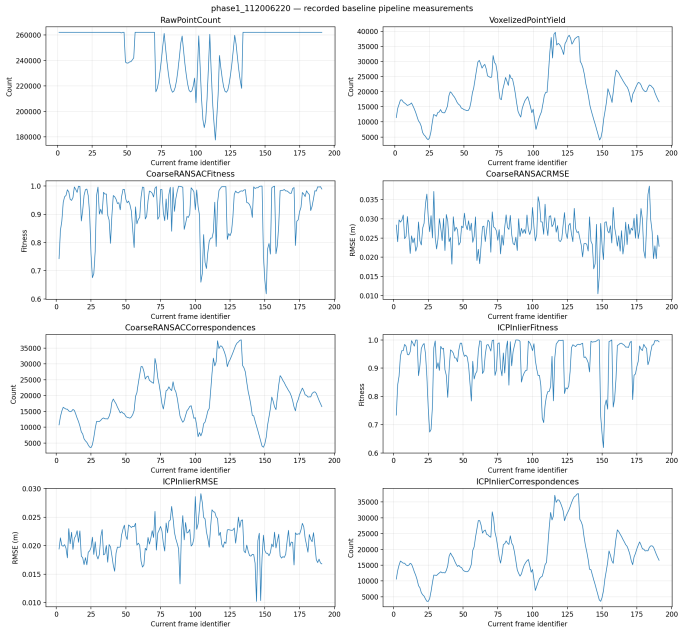


Figure 1. Eight recorded Phase 1 factors

Selection reduced elapsed time by 11.6% but raised aggregate Mean L2 by 75.1%. On the same 142 frames, baseline error is 0.058156 m versus 0.103037 m after reconstruction. Thus this intervention worsened accuracy. The original baseline is preferred; increased motion across omitted frames is a plausible, unproven contributor.

Additional diagnostic trials also failed to improve Phase 1: finer ICP 0.060483 m, temporal ICP 0.241795 m, and 1 cm Huber loss 0.063683 m versus baseline 0.058835 m. They remain diagnostic pilots, not replacements for the sealed baseline. Broad room layout is visible, but no quantitative completeness metric was measured.

Phase 2: Selection and Registration Treatment

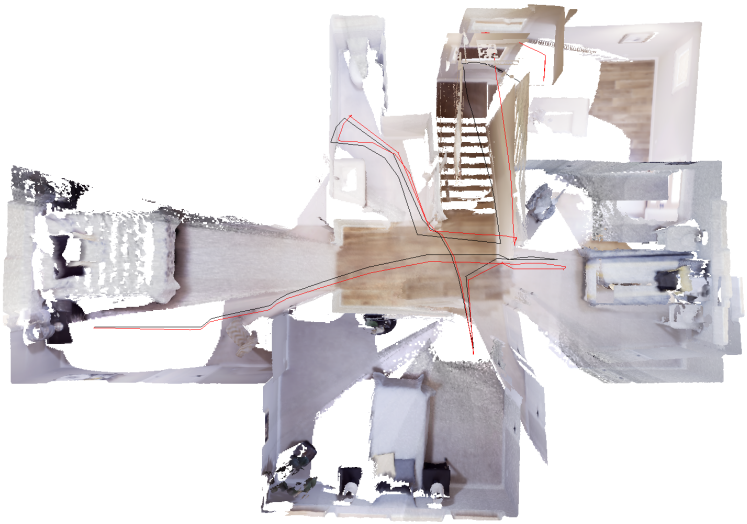
The original student Route03 contains 388 views of bedrooms, corridor/landing and stair descent: 474.091 s of collection and 28.712 m sampled travel. The route is not exclusively upstairs. The supplied navmesh disconnects the lower living area from the staircase endpoint, so that area was absent from this capture.

Run	Frames	Mean L2 (m)	Elapsed (s)	Profile
Original baseline	388	2.092883	278.694	standard
Selected upstairs	262	0.365944	215.514	standard
Same full capture	388	0.390671	471.559	temporal

The selected run keeps RGB-inspected upstairs frames through 270 with VoxelizedPointYield >= 2000, near the observed sparse tail. It retains 262 frames and omits the difficult descent. On these identical 262 frames, original baseline poses score 0.254445 m and selected reconstruction 0.365944 m. Its lower full-run score therefore reflects changed scope, not improved alignment.



Original standard: 388 frames



Temporal + surface: 388 frames

Temporal multi-start ICP: a controlled improvement

The temporal treatment retains the identical full 388-frame input. Each pair still runs RANSAC, then Open3D point-to-plane ICP from RANSAC, identity and previous estimated-motion initializations. Non-finite, zero-support, >1 m translation and >60 degree rotation candidates are rejected. Among candidates within 0.01 of best fitness, choose minimum RMSE + 0.005 times translation/rotation change from previous motion. GT is not used.

Mean L2 falls from 2.092883 to 0.390671 m, an 81.3% same-frame reduction. On the same upstairs 262-frame subset, temporal poses score 0.209269 m versus standard 0.254445 m. All 3098 factors are recorded; 141/114/132 pairs choose RANSAC/identity/previous-motion starts, with no held-motion fallback. Coarse-stage measurements match the original baseline. This supports this treatment on this capture, not a guarantee for other routes.

The images also differ in representation: the original is merged points; the temporal image uses estimated-pose TSDF. Smoothing and camera changes must not be attributed entirely to more accurate poses. Neither fusion nor visualization changes the measured trajectory.

New Captures: Coverage Does Not Ensure Accuracy

After inspecting incomplete results, the student collected a longer upstairs route (Route04) and a separate downstairs visit. Both saved normally with matched RGB/depth/GT counts and finite poses. They remain separate coordinate systems and are reconstructed separately; no GT alignment is used to combine floors.

Capture	Frames	Mean L2 (m)	Elapsed (s)	Profile
Upstairs Route04	738	1.391594	638.368	temporal
Extra downstairs	704	1.252309	1636.978	temporal

Route04 spans 37.577 m sampled travel and camera heights 0.725-2.925 m, including stair descent. The downstairs visit spans 54.820 m and heights 0.125-0.301 m, with living, dining and kitchen areas visible in sampled images. These observations establish visited areas, not full-floor completeness. Downstairs was reached using a separate navigable spawn, not by fixing the disconnected stairs.

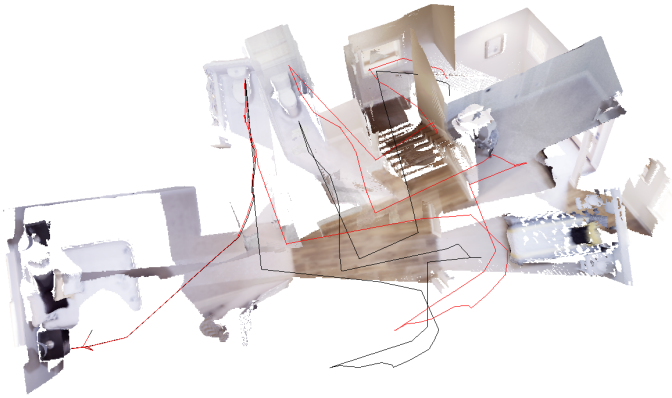


Figure 2. New upstairs: 738 frames



Figure 3. Additional downstairs: 704 frames

Both surfaces use estimated poses, 2 cm TSDF voxels, 8 cm truncation and a ceiling crop. Red/black paths diverge and walls/floors overlap or shift. These are registration artifacts, not merely a tilted view. The longer upstairs route scores worse than the earlier temporal route (1.391594 vs 0.390671 m), but different frames and coverage prevent a controlled causal comparison. The new downstairs data do not replace the required supplied first-floor experiment.

Failure Analysis and Verification Evidence

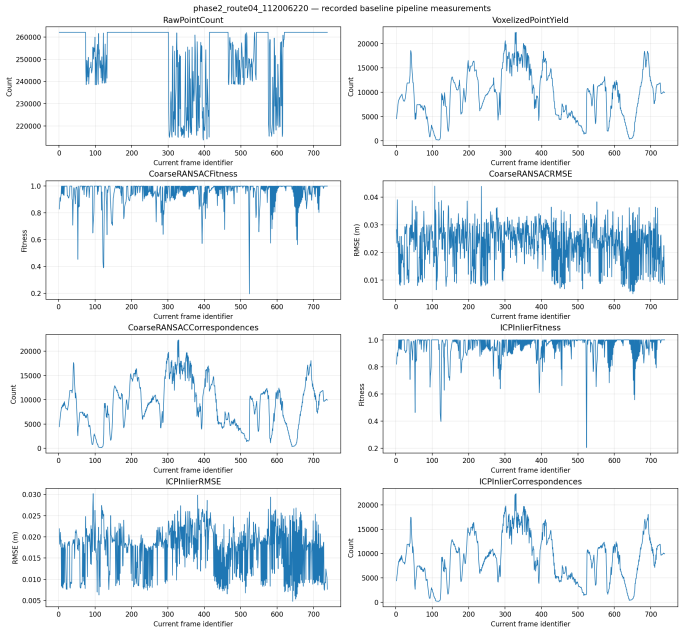


Figure 4. New upstairs: all eight baseline factors

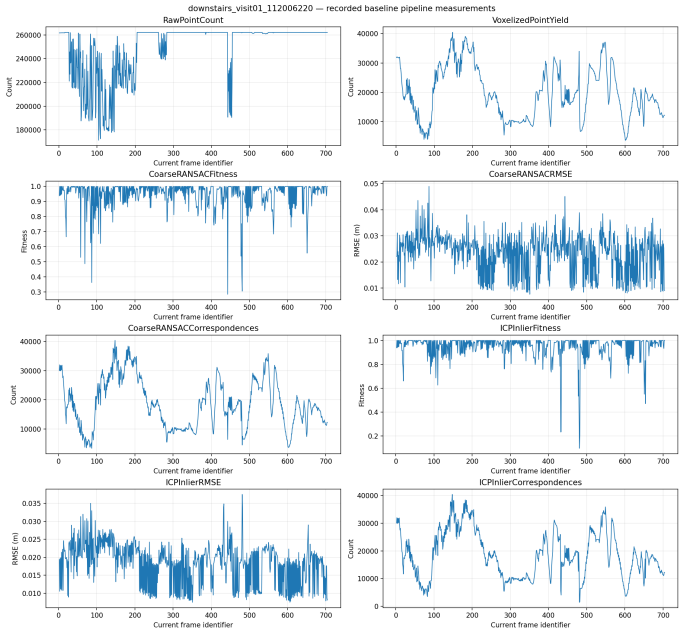


Figure 5. Downstairs: all eight baseline factors

Why the map differs from the handout

The handout illustrates a coherent floor layout viewed from above, with estimated and GT paths close together. Our new maps contain displaced copies of surfaces. Sequential reconstruction composes local transforms; a wrong pair alignment can shift or rotate the following observations. Revisiting a room then adds inconsistent geometry instead of automatically correcting earlier poses. The implemented pipeline has no loop-closure pose-graph optimization.

Upstairs frame range	Mean position error (m)
1-100	0.023648

101-200	1.313437
201-300	2.631858
301-400	2.024992

The first 100 upstairs frames average about 2.4 cm error, but frames 201-300 average 2.63 m. Failure therefore starts before the final stair section. This GT-based breakdown is post-run evaluation only; it did not guide selection or enter reconstruction. Detailed values are preserved in trajectory_error_blocks.json.

Sampled RGB views include dark and saturated regions under the configured spatial perturbations, and logs contain low-mutual-correspondence warnings. These observations motivate checking local geometry and matching support, but do not prove that exposure caused the depth-registration failure. Missing surfaces may reflect unobserved areas, invalid depth, misalignment or cropping. Their individual contributions were not isolated.

The original Route03 evidence provides a clear caution: frame 293 has zero ICP correspondences and zero inlier RMSE; very sparse views have as few as 162 voxelized points. High median fitness (0.96916) did not prevent its full-route drift. The new factor plots likewise remain diagnostic measurements rather than qualification labels.

Discussion and Required Questions

Cross-phase findings

The supplied first-floor baseline is the most accurate measured full capture (0.058835 m). Both original frame-selection interventions worsened matched-frame accuracy. Temporal multi-start ICP improved the original upstairs route on identical frames, but did not prevent drift in the longer captures. More frames increased observed coverage and computation without guaranteeing better geometry. Floors, route scope, perturbations and view sequences differ, so cross-capture scores are descriptive, not controlled ablations.

Q1. What happens without global registration?

ICP is local. With sufficiently small motion and overlap, identity initialization may work; with larger rotations, gaps or repeated structures, nearest-neighbor matches can converge to an incorrect local solution or lack support. FPFH/RANSAC provides a broader initial alignment [1,2]. No standalone ICP-only ablation was measured. The temporal method still runs RANSAC and compares multiple ICP starts; its identity choices are not an ICP-only experiment.

Q2. Which choices improve ICP stability?

Implemented choices include 5 cm voxels, normals within 10 cm, a 7.5 cm correspondence gate and bounded point-to-plane ICP. The temporal treatment adds alternative initializations and motion/support checks. Its measured benefit is specific to the original 388-frame capture. A 1 cm Huber pilot did not improve Phase 1 and is not adopted. Multi-scale ICP and loop-closure optimization are future experiments, not completed improvements.

Q3. How do factors and collection choices relate to results?

Raw/voxelized counts quantify available support; fitness and correspondence counts indicate pairwise matching under a specified gate; inlier RMSE describes accepted residuals. None establishes global correctness alone. Intermediate overlapping views can ease matching, while close-wall, weak-geometry, large-turn and sparse-depth views can be difficult. More frames may add useful rooms or redundant and difficult observations. Matched-frame comparisons distinguish real registration improvements from dropping difficult route sections.

Limitations and next experiments

Coverage is assessed visually, not with a quantitative map-completeness metric. Routes include stair views and are not strictly upstairs-only. New captures currently have baseline results only. Timings are single runs with shared machine load, not isolated performance benchmarks. The custom ICP bonus is not attempted. No zero-error or reference-quality reconstruction is claimed.

Next steps are to locate weak pairs using recorded factors and input views, test justified selection or registration changes under new declarations, and compare identical frames plus visible coverage. Loop closures could address revisit inconsistency, but require implementation and validation. Current results do not justify replacing the earlier lower-error upstairs run simply because the new capture is longer.

Reproducibility, Evidence, and Sources

Environment and validation

pixi.lock specifies Python 3.9.23, NumPy 1.26.4, Open3D 0.18.0, Habitat-Sim/Lab 0.3.3 and RDFLib 7.1.4. Mesa software rendering is used on WSL. The last implementation suite passed 48/48 tests (40 unit/import/capture-control and 8 simulator integration); this report revision does not claim a new full-suite execution. Checks cover manual geometry, coordinate/pose composition, subset matching, ICP rejection and pose exports. A short identical-seed repeat matched trajectories; no full fresh-machine reproduction was performed.

Elapsed time in tables runs from CLI argument processing through reconstruction and image generation. It excludes interpreter imports, final artifact serialization, Turtle append, later TSDF fusion and interactive inspection. This timer limitation applies to all comparisons. The newer runs took 638.368 s upstairs and 1636.978 s downstairs; system load and three ICP starts affect cost.

Local evidence and replay commands

experiments/phase1_112006220/: sealed Turtle, metrics_baseline.json, map/trajectory artifacts, plots/factors_static.png and run_comparison.csv.

experiments/phase2_112006220/: sealed Turtle, metrics_baseline.json, map/trajectory artifacts, plots/factors_static.png and run_comparison.csv.

experiments/phase2_temporal_112006220/: sealed Turtle, metrics_baseline.json, map/trajectory artifacts, plots/factors_static.png and run_comparison.csv.

experiments/phase2_route04_112006220/: sealed Turtle, metrics_baseline.json, map/trajectory artifacts, plots/factors_static.png and run_comparison.csv.

experiments/downstairs_visit01_112006220/: sealed Turtle, metrics_baseline.json, map/trajectory artifacts, plots/factors_static.png and run_comparison.csv.

Baseline collection/registration settings are recorded in each declaration. An unsealed declaration is required for a new measured treatment. Re-running a sealed experiment without --write creates preview artifacts and does not overwrite its recorded evidence. Original selection queries are stored under the original experiments/queries/frames.rq.

Inspect new maps: pixi run -e habitat view-map experiments/phase2_route04_112006220 --surface ; use experiments/downstairs_visit01_112006220 for downstairs. Rebuild report: pixi run -e habitat python scripts/build_report.py --final. See docs/improving-reconstruction.md for temporal settings and TSDF reproduction.

Sources

[1] Open3D 0.18: Global registration. https://www.open3d.org/docs/0.18.0/tutorial/pipelines/global_registration.html

[2] Open3D 0.18: ICP registration. https://www.open3d.org/docs/0.18.0/tutorial/pipelines/icp_registration.html

[3] Open3D 0.18: RegistrationResult. https://www.open3d.org/docs/0.18.0/python_api/open3d.pipelines.registration.RegistrationResult.html

[4] Open3D 0.18: RGB-D integration. https://www.open3d.org/docs/0.18.0/tutorial/pipelines/rgbd_integration.html

[5] Course-provided Phase 1 archive. https://drive.google.com/file/d/17dnyifA0LOi6CXWGOYgl3db1WSuOOHbl/view

[6] Assignment handout: HW1, Sections 2-4 (pipeline, data verification, assessment and report format).